2.4 Q2 → iii

⇒

iii)
$$\begin{vmatrix} k & b & a & c+d \\ k & b & c & a+d \\ k & d & c & b+a \\ k & a & d & b+c \end{vmatrix} = 0$$

sol adding column C_2 and C_3 in C_4.

$$\begin{vmatrix} k & b & a & b+a+c+d \\ k & b & c & b+c+a+d \\ k & d & c & d+c+b+a \\ k & a & d & a+d+b+c \end{vmatrix} = 0$$

now, taking common :- $(a+b+c+d)$ from C_4.
and k common for C_1

$$k(a+b+c+d) \begin{vmatrix} 1 & b & a & 1 \\ 1 & b & c & 1 \\ 1 & d & c & 1 \\ 1 & a & d & 1 \end{vmatrix}$$

so if the two column are same (ide
the $|A| = 0$

$$k(a+b+c+d)(0) = 0$$
$$0 = 0.$$

→ if two rows or two columns are same
then the $|A| = 0$.

2.4

Q3:- without expanding determinants, prove that.

i) $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$

add c_2 in c_3

$\begin{vmatrix} 1 & a & a+b+c \\ 1 & b & b+c+a \\ 1 & c & a+b+c \end{vmatrix} = 0$

$(a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0$

$(a+b+c)(0) = 0$

$\boxed{0 = 0}$

ii) ⟹ $$\begin{vmatrix} x+1 & x+3 & x+5 \\ x+4 & x+6 & x+8 \\ x+7 & x+9 & x+11 \end{vmatrix} = 0$$

sol

subtract $c_2 - c_1$ and $c_3 - c_1$

$$\begin{vmatrix} x+1 & x+3-x-1 & x+5-x-1 \\ x+4 & x+6-x-4 & x+8-x-4 \\ x+7 & x+9-x-7 & x+11-x-7 \end{vmatrix}$$

$$\begin{vmatrix} x+1 & 2 & 4 \\ x+4 & 2 & 4 \\ x+7 & 2 & 4 \end{vmatrix} = 0$$

Now sub $c_3 - c_2$

$$\begin{vmatrix} x+1 & 2 & 4-2 \\ x+4 & 2 & 4-2 \\ x+7 & 2 & 4-2 \end{vmatrix} = 0$$

or take ~~rotum~~ comnon 2 from colum 2 and 4 from 3

$$\begin{vmatrix} x+1 & 2 & 2 \\ x+4 & 2 & 2 \\ x+7 & 2 & 2 \end{vmatrix} = 0$$

Hence two colums are identical

$|A| = 0.$

$2(0) = 0.$

Exercise 2.5

=> Reduce the following

$$\rightarrow iv \quad \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

sol sub R_2 and R_3 with R_1

$$\begin{vmatrix} b+c(-b-c) & a(+c+a)c & a-b-a-b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

Take -2 common from R_1

$$-2\begin{vmatrix} 0 & +c & b \\ b & c+a & b \\ c & c & a+3 \end{vmatrix} = 4abc$$

$\rightarrow R_2$ ko R_1 sai Minus

sub. $R_2 - R_1$ and $R_3 - R_1$

$$-2\begin{vmatrix} 0 & c & b \\ b-0 & c+a-c & b-b \\ c-a & c-c & a+b-b \end{vmatrix} = 4abc$$

$$-2\begin{vmatrix} 0 & c & b \\ b & a & 0 \\ c & 0 & a \end{vmatrix} = 4abc$$

Now after findindig Det you will get

$4abc = 4abc.$